Enhancing Trait Mindfulness in Military Parents Through Online Intervention

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Introduction

Background. Wartime deployment has affected more than 3 million family members in the U.S. since 2001. Military parents who were deployed are at high risk for mental health problems such as depression, anxiety, and PTSD. Their civilian spouses show similar rates of mental health problems (Eaton et al., 2008). Moreover, children in post-deployment families encounter more emotional difficulties in their lives than do their civilian counterparts (Chandra et al., 2010).

While a variety of preventive intervention programs have been applied, mindfulness training has emerged as an effective approach for building resilience, particularly for military service members (Jha, Stanley, Kiyonaga, Wong, & Gelfand, 2010). Mindfulness training has also drawn increased attention in parenting training programs that have primarily used traditional face-to-face group intervention. Compared with traditional face-to-face approaches, Internet-based interventions are merited because they can be self-directed and are easily accessible. While such studies are lacking, Internet-based mindfulness training has showed promises in feasibility and effectiveness (Boettcher et al., 2014; Van Gemert-Pijnen et al., 2014).

Aims. This study drew on data from a larger intervention project within the context of a parenting training program designed for post-deployed military families (After Deployment: Adaptive Parenting Tools). Using longitudinal data, we sought to investigate the characteristics of online mindfulness exercises (OME) usage among parents. Moreover, it examined whether OME usage would significantly predict mothers' and fathers' trait mindfulness at 6-month follow-up assessment, after controlling for deployment status, group attendance, and baseline trait mindfulness.

Week	ADAPT session	OME
0	Building Blocks of Resilience	None
1	Encouraging Cooperation	Mindful eating & Ten deep breaths
2	Teaching New Behavior	Sitting and observing
3	Recognizing Emotions	Body scan & Notice five things
4	Responding to Emotions	Drop Anchor, Leaves on a stream &Bell Ringing
5	Setting Limits	Sitting and observing
6	Follow Through	Hexaflexercise
7	Communicating with Children	Hexaflexercise
8	Problem Solving	Wisdom
9	Managing Conflict	Loving kindness
10	Monitoring Children	Stretching 1
11	Promoting School Success	Stretching 2
12	Linking Home and School	Loving kindness
13	Balancing Work and Play	None

Methods

Sampling. The sample consisted of 370 parents (n=207 families; 51% female, M_{age}=36.79, SD = 6.04). Most of the male participants (96%) had been deployed whereas a majority of the female (79%) participants had not been deployed.

Procedure. An evidence-based parenting training program was provided to all participants, and included 14 weekly face-to-face group sessions as well as Internet-based resources designed to enhance the training. Particularly, 13 brief exercises (audio or video formatted downloadable files) such as "mindful eating", "sitting and observing", and "loving kindness" were embedded in the exercises. These OME were introduced progressively in group sessions and assigned as home practice.

Measures. OME use was logged. Specifically, use referred to the event when an OME was clicked. For each time of use, the exercise component and date stamp were recorded. The Five Facets Mindfulness Questionnaire (FFMQ, Baer et al., 2006) was administered at baseline and at 6-month follow-up assessments. A composite score was used, where the higher score indicates higher trait mindfulness.

Results

- 2,423 individual clicks of OME use were observed.
- Each component was used by approximately 1/3 of the participants.
- Usage over time showed different patterns for the individuals who attended to groups and individuals who participated in the online-only groups. Specifically, the individuals who attended face-to-face groups (n=149) used OME throughout the intervention period, whereas the online-only individuals (n=16) used OME during the first three weeks of the 14-week program.
- OME usage was significantly higher among female individuals (M = 7.93, SD = 10.55) than male individuals (M = 5.09, SD = 8.49), $F(1)$ =4.03, p < .05. However, the gender difference were no longer significant when taking group attendance into account.
- For women, adding usage into the hierarchical regression model explained an additional 3.2% of the variance in trait mindfulness at trait mindfulness at follow-up beyond deployment status, trait mindfulness at baseline, and group attendance (Table 1). On the contrary, the predictive value of OME usage was not statistically significant for men (Table 2).

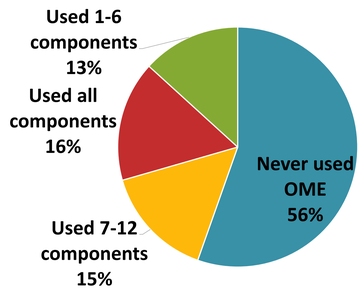


Figure 1. Percentages of Participants with different Levels of usage

Table 1. Summary of hierarchical regression analysis for variables predicting mindfulness at follow-up in women who used OME

Variables	B	SEB	β	R	R^2	Δ R^2
Step 1				.17	.03	.03
Deployment status	8.23	5.54	.17			
Group attendance	.12	8.89	.00			
Step 2				.73	.53	.50
Deployment status	-4.86	4.15	-.10			
Group attendance	1.43	6.22	.02			
Mindfulness (Baseline)	.84	.09	.76***			
Step 3				.75	.56	.03
Deployment status	-5.64	4.05	-.12			
Group attendance	2.54	6.07	.03			
Mindfulness (Baseline)	.84	.09	.76***			
OME usage	.32	.14	.18*			

Note: n =78; *p < .05;***p < .001.

Table 2. Summary of hierarchical regression analysis for variables predicting mindfulness at follow-up in men who used OME

Variables	B	SEB	β	R	R^2	Δ R^2
Step 1				.07	.01	.01
Group attendance	-6.28	12.38	-.07			
Step 2				.79	.62	.62
Group attendance	-2.03	7.74	-.02			
Mindfulness (Baseline)	1.05	.12	.79***			
Step 3				.79	.62	.00
Group attendance	-2.40	8.28	-.03			
Mindfulness (Baseline)	1.05	.12	.78			
OME usage	-.03	.22	-.01			

Note: n = 51; ***p < .001.

Implementation

- This study evidenced the feasibility and effectiveness of OME for enhancing parent trait mindfulness.
- It is advisable to consider gender differences and implement supportive strategies to promote usage behaviors when incorporating online mindfulness training in parent-based intervention programs.